

ReferPeople.in

Complete Platform Guide

Deployment · Operations · User Management

Version 1.0 | May 2026 | Server: 13.201.38.123

26	12	12	13
Containers	Backends	Frontends	Databases

Table of Contents

1. What is ReferPeople.in?
2. Platform Architecture
3. All Services — Ports, URLs & What They Do
4. Complete Fresh Deployment Guide
5. Environment Files (.env) Setup
6. Docker Build & Start All Containers
7. Database Setup & Migrations
8. Testing All Connections
9. Super Admin — Access & Operations
10. Admin Account — Create & Manage
11. User Account — Create & Manage
12. Property & Loan Data Management
13. GitHub CI/CD Pipeline
14. SSL & Domain Setup
15. Common Issues & Fixes
16. All Configuration Files Reference

1. What is ReferPeople.in?

ReferPeople.in is a **multi-tenant loan and property management platform** built for financial institutions. It allows organisations to manage the full lifecycle of loans and property valuations through a suite of specialised dashboards.

What the Platform Does

Dashboard / Role	What It Does
Master Admin	Manages the entire platform — all tenants, all users, all products
Tenant Admin	Manages their organisation's staff, branches, and loan products
CRM	Handles customer leads and follow-ups for loan enquiries
Sales CRM	Tracks the sales pipeline from lead to disbursement
Finance Dashboard	Monitors loan disbursements, repayments, and financial reports
Agents App	For field agents who collect documents and verify customers
Channel Partner	For referral partners who bring in loan leads
Fraud Team	Flags and investigates suspicious applications
Legal Dashboard	Manages legal documentation and agreements
Operations	Verifies documents and disburses approved loans
Valuation Dashboard	Handles property valuation for loan collateral
Website	Public-facing landing page for loan enquiries

How a Loan Works (End to End)

- Step 1:** Customer visits referpeople.in and fills enquiry form → Lead created in CRM
- Step 2:** Sales CRM agent calls customer, qualifies them for a loan product
- Step 3:** CRM creates formal loan application with customer documents
- Step 4:** Field Agent collects KYC documents on the ground
- Step 5:** Operations team verifies all documents submitted
- Step 6:** Valuation team visits property (for secured/property loans)
- Step 7:** Legal team prepares the loan agreement
- Step 8:** Finance team approves disbursement and releases funds
- Step 9:** Loan enters repayment tracking in the Finance Dashboard

How Property Valuation Works

- Loan application is flagged as requiring property valuation

- Valuation Dashboard receives the new assignment automatically
- Valuation officer visits the property and fills the inspection report
- Report is uploaded to the Valuation Dashboard with photos and estimated value
- Status updates automatically on the linked loan application
- Legal and Finance teams can now proceed with disbursement

2. Platform Architecture

Infrastructure Components

Component	Service	Details
Web Server	EC2 t3.large	Ubuntu 22.04, 8 vCPU, 8GB RAM, 30GB disk, ap-south-1
Database	RDS PostgreSQL 16	db.t3.micro, Single-AZ, ap-south-1
Cache / Queue	Redis 7	Docker container, port 6379
Proxy	Nginx 1.25	Docker container, routes all HTTP traffic on port 80/443
Runtime	Docker Compose	26 containers total — 12 backends + 12 frontends + nginx + redis

Traffic Flow

All external traffic arrives on port 80 (or 443 with SSL) and Nginx routes it to the correct container based on the URL path. Backends expose internal Django REST APIs; frontends serve compiled React apps.

URL Path	Container	Port	Purpose
/	website-frontend	3000	Public website
/master-admin/	masteradmin-frontend	3001	Platform super-admin panel
/api/master-admin/	masteradmin-backend	8001	Django REST API
/tenant/	tenant-frontend	3002	Tenant portal
/api/tenant/	tenant-backend	8002	Django REST API
/crm/	crm-frontend	3003	CRM dashboard
/api/crm/	crm-backend	8003	Django REST API
... (all 12 pairs)	...	...	Same pattern for all services

3. All Services — Ports, URLs & What They Do

Frontend Services

Container	Port	URL	Description
website-frontend	3000	http://13.201.38.123/	Public website / loan enquiry form
masteradmin-frontend	3001	http://13.201.38.123/master-admin/	Platform super-admin panel
tenant-frontend	3002	http://13.201.38.123/tenant/	Tenant organisation portal
crm-frontend	3003	http://13.201.38.123/crm/	CRM — leads & customers
finance-frontend	3004	http://13.201.38.123/finance/	Finance & disbursement
salescrm-frontend	3005	http://13.201.38.123/sales-crm/	Sales pipeline
tenantadmin-frontend	3006	http://13.201.38.123/tenant-admin/	Tenant admin panel
channel-frontend	3007	http://13.201.38.123/channel/	Channel partners
fraud-frontend	3008	http://13.201.38.123/fraud/	Fraud detection team
legal-frontend	3009	http://13.201.38.123/legal/	Legal team dashboard
operations-frontend	3010	http://13.201.38.123/operations/	Operations verification
valuation-frontend	3011	http://13.201.38.123/valuation/	Property valuation

Backend Services (Django REST APIs)

Container	Port	API Route	Django Admin
masteradmin-backend	8001	/api/master-admin/	:8001/admin/
tenant-backend	8002	/api/tenant/	:8002/admin/
crm-backend	8003	/api/crm/	:8003/admin/
finance-backend	8004	/api/finance/	:8004/admin/
agents-backend	8005	/api/agents/	:8005/admin/
channel-backend	8006	/api/channel/	:8006/admin/
fraud-backend	8007	/api/fraud/	:8007/admin/
legal-backend	8008	/api/legal/	:8008/admin/
operations-backend	8009	/api/operations/	:8009/admin/
salescrm-backend	8010	/api/sales-crm/	:8010/admin/

tenantadmin-backend	8011	/api/tenant-admin/	:8011/admin/
valuation-backend	8012	/api/valuation/	:8012/admin/

Databases on RDS

Database Name	Used By	Contents
masteradmin_db	masteradmin-backend	Core platform users and tenants
tenant_db	tenant-backend	Tenant organisation data
crm_db	crm-backend	Leads, contacts, loan applications
finance_db	finance-backend	Disbursements and repayments
agents_db	agents-backend	Field agent activities
channel_db	channel-backend	Channel partner referrals
fraud_db	fraud-backend	Fraud flags and investigations
legal_db	legal-backend	Agreements and legal docs
operations_db	operations-backend	Document verification tasks
salescrm_db	salescrm-backend	Sales pipeline and targets
tenantadmin_db	tenantadmin-backend	Tenant admin configuration
valuation_db	valuation-backend	Property valuation reports

4. Complete Fresh Deployment Guide

This section covers deploying from scratch on a brand-new AWS server. Follow every step in order.

4.1 AWS Prerequisites

- AWS account with billing enabled
- Domain name (referpeople.in) with DNS access
- GitHub access to the BRD_FINAL repository
- EC2 key pair (.pem file) saved on your laptop

4.2 Create VPC & Subnets

Go to **AWS Console** → **VPC** → **Create VPC**

VPC name: referpeople-vpc
IPv4 CIDR: 10.0.0.0/16

Subnet 1 (Public):

Name: referpeople-public-1a | AZ: ap-south-1a | CIDR: 10.0.1.0/24

Subnet 2 (Public):

Name: referpeople-public-1b | AZ: ap-south-1b | CIDR: 10.0.2.0/24

Internet Gateway: referpeople-igw → attach to referpeople-vpc

Route Table: add 0.0.0.0/0 → referpeople-igw

4.3 Create RDS PostgreSQL Database

Go to **AWS Console** → **RDS** → **Create database**

Engine: PostgreSQL 16
Template: Free tier
DB identifier: referpeople-db
Master username: referpeopleadmin
Master password: ReferPeopleDB2024
Instance: db.t3.micro
Storage: 20 GB gp2
VPC: referpeople-vpc
Public access: No
Security group: Allow port 5432 from EC2 security group

4.4 Create EC2 Instance

Go to **AWS Console** → **EC2** → **Launch Instance**


```
Name: referpeople-server
AMI: Ubuntu Server 22.04 LTS
Instance type: t3.large (8GB RAM – REQUIRED for 26 containers)
Key pair: referpeople-key (create new, download .pem file)
VPC: referpeople-vpc
Subnet: referpeople-public-1a
Auto-assign IP: Enable
Security group: Port 22 (SSH), Port 80 (HTTP), Port 443 (HTTPS)
Storage: 30 GB gp3 (IMPORTANT: change from default 8GB to 30GB)
```

After launch, go to EC2 → Elastic IPs → Allocate → Associate with your instance. This gives you a permanent IP that never changes.

4.5 Connect to EC2

```
# On your laptop (Mac/Linux):
chmod 400 ~/Downloads/referpeople-key.pem
ssh -i ~/Downloads/referpeople-key.pem ubuntu@13.201.38.123
```

4.6 Install Docker

```
sudo apt update && sudo apt upgrade -y
curl -fsSL https://get.docker.com | sudo bash
sudo usermod -aG docker ubuntu
sudo apt install -y docker-compose-plugin
exit # disconnect and reconnect for group change
ssh -i ~/Downloads/referpeople-key.pem ubuntu@13.201.38.123
docker --version && docker compose version # verify
```

4.7 Add Swap Space (Prevents build crashes)

```
sudo fallocate -l 4G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
free -h # verify: Swap shows 4.0G
```

4.8 Clone Repository

```
cd ~
git clone https://github.com/vizz-bob/BRD_FINAL.git
cd BRD_FINAL
ls # verify all backend and frontend folders are present
```

4.9 Set Up Environment Files

```
nano create-env-files.sh
# Change these two lines at the top:
# DB_PASSWORD="ReferPeopleDB2024" <- your RDS password
# SERVER_IP="13.201.38.123" <- your EC2 Elastic IP

bash create-env-files.sh
find . -name ".env" | wc -l # should show 13 or more
```

4.10 Build and Start Containers

IMPORTANT: Use tmux so SSH disconnects don't kill the build. Builds take 20-40 minutes.

```
sudo apt install -y tmux
tmux new -s deploy

# Build in batches (prevents RAM exhaustion)
docker compose build --no-cache masteradmin-backend masteradmin-frontend tenant-backend
tenant-frontend
docker compose build --no-cache crm-backend crm-frontend finance-backend
finance-frontend
docker compose build --no-cache salescrm-backend salescrm-frontend tenantadmin-backend
tenantadmin-frontend
docker compose build --no-cache agents-backend channel-backend channel-frontend
docker compose build --no-cache fraud-backend fraud-frontend legal-backend
legal-frontend
docker compose build --no-cache operations-backend operations-frontend valuation-backend
valuation-frontend
docker compose build --no-cache website-frontend

# Start everything
docker compose up -d
docker compose ps # all 26 should show 'Up'
```

5. Environment Files (.env) Setup

Each backend reads its configuration from a **.env** file. These files contain database credentials, secret keys, and settings. They are **never committed to git** — they stay only on the server.

Standard Template

```
DEBUG=False
DJANGO_ENV=production
SECRET_KEY=your-secret-key-here
ALLOWED_HOSTS=13.201.38.123, referpeople.in, www.referpeople.in, localhost

DB_NAME=masteradmin_db # change this per backend
DB_USER=referpeopleadmin
DB_PASSWORD=ReferPeopleDB2024
DB_HOST=referpeople-db.cju28y6c0s6d.ap-south-1.rds.amazonaws.com
DB_PORT=5432
PGSSLMODE=require # REQUIRED for AWS RDS

CORS_ALLOWED_ORIGINS=https://referpeople.in, http://13.201.38.123
```

DB_NAME Per Backend

Backend Folder	DB_NAME to use
BRD_MasterAdmin_Backend_1.1	masteradmin_db
BRD-MergedTenantMaster-Backend	tenant_db
BRD_CRM_1.1_BACKEND	crm_db
BRD_FINANCE_DASHBOARD_Backend	finance_db
BRD-AgentsApp-Backend	agents_db
BRD-ChannelPartnerDashboard-Backend	channel_db
BRD-FraudTeam-Dashboard-Backend	fraud_db
BRD-LegalDashboard-Backend	legal_db
BRD-OperationVerification-Backend	operations_db
BRD-SalesCRM-Dashboard-Backend	salescrm_db
BRD-TenantAdmin_backend_2.0	tenantadmin_db
BRD-Valuation-Dashboard-Backend	valuation_db

6. Docker Build & Start All Containers

Essential Commands

Task	Command
Start all containers	<code>docker compose up -d</code>
Stop all containers	<code>docker compose down</code>
Check status of all	<code>docker compose ps</code>
Restart one container	<code>docker compose restart masteradmin-backend</code>
Rebuild + restart one	<code>docker compose up -d --build masteradmin-backend</code>
View live logs	<code>docker compose logs -f masteradmin-backend</code>
View last 50 log lines	<code>docker compose logs --tail=50 masteradmin-backend</code>
Check memory usage	<code>docker stats --no-stream</code>

Expected Output After Start

After **docker compose up -d**, run **docker compose ps**. You should see all 26 containers with status **Up**. None should show **Restarting**.

```
loancrm_masteradmin_backend Up 0.0.0.0:8001->8000/tcp
loancrm_masteradmin_frontend Up 0.0.0.0:3001->80/tcp
loancrm_tenant_backend Up 0.0.0.0:8002->8000/tcp
... (all 26 containers showing Up)
loancrm_nginx Up 0.0.0.0:80->80/tcp
loancrm_redis Up (healthy)
```

7. Database Setup & Migrations

Create All 12 Databases (First Time Only)

```
docker compose exec -T masteradmin-backend python -c "  
import psycopg2  
conn = psycopg2.connect(  
    host='referpeople-db.cju28y6c0s6d.ap-south-1.rds.amazonaws.com',  
    port=5432, user='referpeopleadmin', password='ReferPeopleDB2024',  
    dbname='postgres', sslmode='require'  
)  
conn.autocommit = True  
cur = conn.cursor()  
for db in ['masteradmin_db', 'tenant_db', 'crm_db', 'finance_db', 'agents_db',  
    'channel_db', 'fraud_db', 'legal_db', 'operations_db',  
    'salescrm_db', 'tenantadmin_db', 'valuation_db']:  
    try:  
        cur.execute(f'CREATE DATABASE {db}')        print(f'Created: {db}')    except Exception as e:  
        print(f'Skipped {db}: {e}')"
```

Run Migrations on All Backends

```
for svc in masteradmin-backend tenant-backend crm-backend finance-backend \  
agents-backend channel-backend fraud-backend legal-backend \  
operations-backend salescrm-backend tenantadmin-backend valuation-backend; do  
    echo "=== $svc ==="  
    docker compose exec -T $svc python manage.py migrate --noinput 2>&1 | tail -3  
done
```

Create Super Admin

```
docker compose exec masteradmin-backend python manage.py createsuperuser  
# Enter: Email, Username, Password when prompted
```

8. Testing All Connections

Test All 12 Frontend Routes

```
for path in / /master-admin/ /crm/ /finance/ /sales-crm/ /tenant-admin/ \
/channel/ /fraud/ /legal/ /operations/ /valuation/ /tenant/; do
STATUS=$(curl -s -o /dev/null -w "%{http_code}" --max-time 5 http://localhost$path)
echo "$STATUS http://13.201.38.123$path"
done
# All should return 200
```

Test All Backend API Ports

```
for port in 8001 8002 8003 8004 8005 8006 8007 8008 8009 8010 8011 8012; do
STATUS=$(curl -s -o /dev/null -w "%{http_code}" --max-time 5
http://localhost:$port/admin/)
echo "Port $port: $STATUS"
done
# All should return 200 or 302
```

Test Database Connection

```
docker compose exec masteradmin-backend python -c "
from django.db import connection
cursor = connection.cursor()
cursor.execute('SELECT version()')
print('DB OK:', cursor.fetchone()[0][:50])
"
```

Test Redis

```
docker compose exec redis redis-cli ping
# Should return: PONG
```

Check System Resources

```
df -h / # disk space — keep above 5GB free
free -h # memory — 8GB total on t3.large
docker stats --no-stream # per-container CPU and memory
```

9. Super Admin — Access & Operations

The Super Admin has full platform access — create tenants, manage all users, see all data across the entire platform.

Login URLs

Main Super Admin Panel: <http://13.201.38.123/master-admin/>
Django Admin (raw DB): <http://13.201.38.123:8001/admin/>

Create Super Admin (If Not Done)

```
docker compose exec masteradmin-backend python manage.py createsuperuser
# Prompts for: Email, Username, Password
# Recommended: admin@referpeople.com / Admin@1234
```

What Super Admin Can Do

Function	What It Does
Tenant Management	Create new tenant organisations, activate/deactivate them
User Management	View all users across all tenants, reset passwords
Product Management	Define loan products (home loan, personal loan, etc.)
Platform Reports	Platform-wide analytics — applications, disbursements
System Settings	Configure platform-level defaults and notifications
Subscription	Manage tenant subscription plans and billing

Creating a New Tenant Organisation

Step 1: Log in at <http://13.201.38.123/master-admin/>

Step 2: Go to Tenants → Add New Tenant

Step 3: Fill in: Organisation name, contact details, subscription plan

Step 4: Click Create — tenant gets their login credentials

Step 5: Tenant Admin logs in at <http://13.201.38.123/tenant-admin/>

10. Admin Account — Create & Manage

Types of Admins

Role	Scope
Tenant Admin	Manages entire organisation — all staff, branches, products
Branch Manager	Manages a specific branch and its staff
Team Lead	Manages a team of agents or CRM staff

Method 1: Create via Master Admin Panel

- Step 1:** Log in at <http://13.201.38.123/master-admin/>
- Step 2:** Go to Tenants → select the tenant organisation
- Step 3:** Click Manage Users → Add Admin User
- Step 4:** Fill in: Name, Email, Phone, Role = Admin
- Step 5:** Welcome email with credentials is sent automatically

Method 2: Create via Django Shell

```
docker compose exec tenantadmin-backend python manage.py shell -c "
from django.contrib.auth import get_user_model
User = get_user_model()
user = User.objects.create_superuser(
    email='admin@company.com',
    username='companyadmin',
    password='SecurePass@123'
)
print(f'Admin created: {user.email}')
```

Tenant Admin Login

URL: <http://13.201.38.123/tenant-admin/>
 Email: (created above)
 Password: (set during creation)

Reset Admin Password


```
docker compose exec tenantadmin-backend python manage.py shell -c "  
from django.contrib.auth import get_user_model  
User = get_user_model()  
user = User.objects.get(email='admin@company.com')  
user.set_password('NewPassword@123')  
user.save()  
print('Password reset done')  
"
```

11. User Account — Create & Manage

User Login by Dashboard

Role	Login URL	Purpose
CRM Agent	http://13.201.38.123/crm/	Manage leads and customer applications
Sales Agent	http://13.201.38.123/sales-crm/	Manage sales pipeline and targets
Finance Officer	http://13.201.38.123/finance/	Process disbursements and repayments
Field Agent	http://13.201.38.123/	Mobile app for document collection
Channel Partner	http://13.201.38.123/channel/	Submit and track referral leads
Fraud Analyst	http://13.201.38.123/fraud/	Investigate flagged applications
Legal Officer	http://13.201.38.123/legal/	Prepare and manage agreements
Operations Staff	http://13.201.38.123/operations/	Verify documents and disburse
Valuation Officer	http://13.201.38.123/valuation/	Property valuation assignments

Create User via Tenant Admin Panel

Step 1: Log in at <http://13.201.38.123/tenant-admin/>

Step 2: Go to Users → Add User

Step 3: Fill in: Full Name, Email, Phone, Role, Branch

Step 4: Click Create User — user receives email with temporary password

Step 5: User logs in and changes their password on first login

Create User via Django Shell

```
docker compose exec crm-backend python manage.py shell -c "
from django.contrib.auth import get_user_model
User = get_user_model()
user = User.objects.create_user(
    email='agent@referpeople.com',
    username='agent01',
    password='Temp@12345',
    first_name='John', last_name='Smith'
)
user.save()
print(f'Created: {user.email}')
"
```

Update User Data

```
docker compose exec masteradmin-backend python manage.py shell -c "  
from django.contrib.auth import get_user_model  
User = get_user_model()  
user = User.objects.get(email='agent@referpeople.com')  
user.first_name = 'Updated Name'  
user.phone_number = '9876543210'  
user.save()  
print('Updated successfully')  
"
```

Deactivate a User

```
docker compose exec masteradmin-backend python manage.py shell -c "  
from django.contrib.auth import get_user_model  
User = get_user_model()  
user = User.objects.get(email='agent@referpeople.com')  
user.is_active = False  
user.save()  
print('User deactivated')  
"
```

12. Property & Loan Data Management

Loan Application Flow

Stage	Dashboard	What Happens
1. Lead Created	CRM at /crm/	Customer enquiry received, lead logged
2. Lead Qualified	Sales CRM at /sales-crm/	Agent calls, qualifies for loan product
3. Application Filed	CRM at /crm/	Formal application created with customer details
4. Docs Collected	Agents App	Field agent collects KYC/income documents
5. Docs Verified	Operations at /operations/	Staff verifies submitted documents
6. Property Valued	Valuation at /valuation/	Officer visits and values collateral property
7. Legal Agreement	Legal at /legal/	Agreement prepared and signed
8. Disbursement	Finance at /finance/	Funds released to customer account
9. Repayment	Finance at /finance/	EMI tracked, collections monitored

Property Valuation Workflow

Step 1: Loan application is created and flagged as requiring property valuation

Step 2: Valuation Dashboard receives the new assignment automatically

Step 3: Valuation officer logs in at <http://13.201.38.123/valuation/>

Step 4: Officer clicks Assignments → finds the case → clicks Start Inspection

Step 5: Visits the property, fills the inspection report with dimensions and photos

Step 6: Uploads completed report — status auto-updates on the loan application

Step 7: Legal and Finance teams can now proceed with the loan

Access Django Admin for Any Backend

Each backend has a full database admin panel. Use it to view, edit, or delete any record directly.

Backend	Django Admin URL
Master Admin	http://13.201.38.123:8001/admin/
CRM	http://13.201.38.123:8003/admin/
Finance	http://13.201.38.123:8004/admin/
Valuation	http://13.201.38.123:8012/admin/
Operations	http://13.201.38.123:8009/admin/
Legal	http://13.201.38.123:8008/admin/

Tenant Admin	http://13.201.38.123:8011/admin/
Sales CRM	http://13.201.38.123:8010/admin/

Create Superuser for Each Django Admin

```
# Run once per backend to enable /admin/ access
for svc in crm-backend finance-backend agents-backend channel-backend \
fraud-backend legal-backend operations-backend salescrm-backend \
tenantadmin-backend valuation-backend tenant-backend; do
echo "=== $svc ==="
docker compose exec -T $svc python manage.py shell -c "
from django.contrib.auth import get_user_model
User = get_user_model()
if not User.objects.filter(email='admin@referpeople.com').exists():
User.objects.create_superuser(email='admin@referpeople.com',
password='Admin@1234', username='admin')
print('Created')
else:
print('Already exists')" 2>&1
done
```

13. GitHub CI/CD Pipeline

Every push to the **main** branch automatically deploys the latest code to EC2. You never need to SSH in to deploy — just push code.

How the Pipeline Works

Step	When	What It Does
1. Code Checks	Runs on every push	Verifies all folders exist, Python syntax, YAML valid
2. SSH to EC2	Only on main branch	Uses EC2_SSH_KEY secret to connect
3. Sync Code	rsync	Copies changed files to /home/ubuntu/BRD_FINAL/
4. Build Containers	docker compose build	Rebuilds only changed services
5. Run Migrations	manage.py migrate	Applies any new database migrations
6. Health Check	curl	Checks site returns 200 OK

Required GitHub Secrets

Go to: **GitHub Repo** → **Settings** → **Secrets and variables** → **Actions** → **New repository secret**

Secret Name	Value	Description
EC2_HOST	13.201.38.123	Your EC2 Elastic IP address
EC2_USER	ubuntu	SSH username for EC2
EC2_APP_DIR	/home/ubuntu/BRD_FINAL	Path to the app on EC2
EC2_SSH_KEY	Contents of referpeople-key.pem	Full private key including BEGIN/END lines

```
# Get SSH key content (run on your laptop):  
cat ~/Downloads/referpeople-key.pem  
# Copy ALL output including -----BEGIN RSA PRIVATE KEY----- lines  
# Paste as value of EC2_SSH_KEY secret in GitHub
```

Check Pipeline Status

```
# View at: https://github.com/vizz-bob/BRD_FINAL/actions  
  
# Green checkmark = deployed successfully  
# Red X = failed – click to see which step failed and the error logs
```

Manual Deploy (Without Git Push)

```
# SSH into EC2, then:  
cd /home/ubuntu/BRD_FINAL  
git pull origin main  
docker compose up -d --build  
docker compose exec -T masteradmin-backend python manage.py migrate --noinput
```

14. SSL & Domain Setup

Step 1: Point DNS to EC2

Log in to your domain registrar and add these DNS records:

```
Type: A Name: @ Value: 13.201.38.123 TTL: 300
Type: A Name: www Value: 13.201.38.123 TTL: 300
```

Wait 15 minutes, then verify:

```
nslookup referpeople.in
# Should show: Address: 13.201.38.123
```

Step 2: Get SSL Certificate via Certbot

```
sudo apt install -y certbot python3-certbot-nginx
sudo certbot --nginx -d referpeople.in -d www.referpeople.in
# Enter email address when prompted
# Type Y to agree to terms
# Type N to decline EFF newsletter
```

Step 3: Verify HTTPS Works

```
curl -I https://referpeople.in
# Should return: HTTP/2 200
```

Step 4: Confirm Auto-Renewal

```
sudo certbot renew --dry-run
# Should show: Congratulations, all simulated renewals succeeded
```

Certbot automatically creates a cron job that renews the certificate every 60 days. You do not need to manually renew.

15. Common Issues & Fixes

Problem: Container shows Restarting

Fix: `docker compose logs --tail=20` # check the error

- Wrong .env credentials
- Migration error (see below)
- Missing Python package

Problem: DB connection failed — password auth

Fix: Check .env has correct DB_USER, DB_PASSWORD, PGSSLMODE=require

- After fixing .env: `docker compose up -d --force-recreate`

Problem: Migration error — column already exists

Fix: `docker compose run --rm --no-deps python manage.py migrate --fake`

- This skips the conflicting migration safely

Problem: 502 Bad Gateway from Nginx

Fix: `docker compose ps` # check backend is Up

- `docker compose restart nginx`
- Check `nginx/conf.d/loancrm.conf` uses service names

Problem: No space left on device during build

Fix: `docker system prune -af` # removes unused images

- `df -h /` # check disk usage

Problem: EC2 freezes during Docker build

Fix: `free -h` # check if swap is present

- Add 4GB swap (Section 4.7)
- Build in smaller batches of 2-3 services

Problem: SSH disconnects kill the build

Fix: Always use `tmux` for long operations

- `tmux new -s work` → do work → `Ctrl+B D` to detach

Problem: GitHub Actions fails with SSH error

Fix: Re-check EC2_SSH_KEY secret includes BEGIN/END lines

- Make sure EC2 security group allows port 22 from GitHub IPs

16. All Configuration Files Reference

File	Location	Purpose
docker-compose.yml	/home/ubuntu/BRD_FINAL/	Defines all 26 containers. env_file, ports, networks, depends_on.
nginx/conf.d/loancrm.conf	/home/ubuntu/BRD_FINAL/nginx/conf.d/	Routes URL paths to frontend/backend containers via Docker service names.
docker-entrypoint.sh	/home/ubuntu/BRD_FINAL/	Startup script: waits for DB, runs migrate, starts Gunicorn.
.github/workflows/deploy.yml	/home/ubuntu/BRD_FINAL/.github/workflows/	CI/CD: code checks on every push, auto-deploy on push to main.
create-env-files.sh	/home/ubuntu/BRD_FINAL/	Script to generate all 13 .env files. Edit DB_PASSWORD and SERVER_IP first.
*/requirements.txt	Each backend folder	Python packages for each backend. Add missing packages here and rebuild.
*/Dockerfile	Each backend/frontend folder	Docker build instructions for each service.

Quick Reference Card

Daily Operations

Task	Command
Check all containers	docker compose ps
View live logs	docker compose logs -f
Restart one container	docker compose restart
Restart everything	docker compose down && docker compose up -d
Check disk space	df -h /
Check memory	free -h
Remove unused Docker data	docker system prune -af

Important URLs

What	URL
Main Website	http://referpeople.in or http://13.201.38.123/
Super Admin Panel	http://13.201.38.123/master-admin/
CRM Dashboard	http://13.201.38.123/crm/

Finance Dashboard	http://13.201.38.123/finance/
Valuation Dashboard	http://13.201.38.123/valuation/
GitHub CI/CD Pipeline	https://github.com/vizz-bob/BRD_FINAL/actions

Key Credentials

What	Value
EC2 IP Address	13.201.38.123
EC2 SSH User	ubuntu
RDS Host	referpeople-db.cju28y6c0s6d.ap-south-1.rds.amazonaws.com
RDS Username	referpeopleadmin
Super Admin Email	admin@referpeople.com
App Directory	/home/ubuntu/BRD_FINAL

ReferPeople.in — Complete Platform Guide v1.0 | Generated May 2026 | Confidential